

Windows Event Viewer User Login and Failed Password Log Analysis Report

1. Introduction

Windows Event Viewer is a built-in Windows tool that records operating system events, security activities, application logs, and authentication events.

This project focuses on monitoring successful and failed user authentication events using the Windows Security Log. Such monitoring is one of the primary responsibilities of a Security Operations Center (SOC) Analyst.

2. Objective

The objectives of this project are:

- Analyze Windows Security Logs**
- Monitor Successful User Login**
- Detect Failed Password Attempts**
- Understand Windows Authentication Events**
- Learn Security Event Investigation**

3. Lab Environment

Operating System:

Windows 10 / Windows 11

Virtualization:

Oracle VirtualBox

Tool Used:

Windows Event Viewer

Log Source:

Windows Security Log

Step 1

Open Event Viewer using:

eventvwr.msc

Step 2

Navigate to:

Windows Logs



Security

Step 3

Apply Log Filter

Event IDs

4624

4625

Step 4

Generate Authentication Events

Successful Login

Login using the correct password.

Failed Login

Attempt login using an incorrect password multiple times.

Step 5

Observe Event Details

Record the following information:

- Event ID
- Date
- Time
- Username

- Logon Type
- Computer Name
- Failure Reason
- Status Code
- Sub Status Code
- Source Network Address

5. Event Analysis

Event ID 4624

Description

Successful user authentication.

Observed Fields

- Username
- Logon Type
- Authentication Package
- Process Name
- Logon Process
- Source Address

SOC Interpretation

The user successfully authenticated into the Windows system.

Event ID 4625

Description

Failed login attempt.

Observed Fields

- Username
- Failure Reason

- **Status Code**
- **Sub Status Code**
- **Logon Type**
- **Source Address**

SOC Interpretation

An authentication attempt failed because the supplied credentials were invalid or another authentication error occurred.

6. Results

The Windows Security Log successfully recorded:

- Successful User Login
- Failed Password Attempts
- User Authentication Details
- Login Time
- Username
- Authentication Information

The Event Viewer proved useful for monitoring authentication events and identifying suspicious login activities.

7. Conclusion

This project demonstrates how Windows Event Viewer can be used to investigate authentication events in a Windows environment. By analyzing Event IDs 4624 and 4625, security analysts can identify legitimate user logins, failed password attempts, and potential brute-force attacks. This practical exercise strengthens foundational SOC monitoring and incident investigation skills.

8. Skills Gained

- Windows Event Viewer
- Security Log Analysis
- Windows Authentication
- Event ID Investigation
- SOC Monitoring
- Incident Analysis
- Authentication Log Investigation
- Basic Digital Forensics